

Task 1: Introduction to React & JSX Basics

- Learned what React is and why it is used.
- Understood JSX syntax. Learned about Components.
- Learned how React updates the UI.
- Read the official React documentation.
- Watched the assigned introductory video.

Summary

React is an open-source JavaScript library developed by Meta (Facebook) for building fast, interactive, and reusable User Interfaces (UI). It follows a component-based architecture, where the user interface is divided into small reusable parts called components. This approach makes the code more organized, reusable, and easier to maintain.

JSX (JavaScript XML) is a syntax extension used in React that allows developers to write HTML-like code inside JavaScript. JSX makes React code more readable and simplifies UI development. Although JSX looks similar to HTML, it is not understood directly by web browsers. It is converted into regular JavaScript by Babel before execution.

A React Component is simply a JavaScript function that returns JSX. Components can be reused multiple times within an application, reducing code duplication and making development more efficient.

JSX basic code

```
function App() {  
  return (  
    <h1>Hello World</h1>  
  );  
}
```

Task 2: Create React App using Vite

Creating a React Project using Vite

To create a React application, Vite provides a fast development environment.

Steps

1. Create project using command in terminal: `npm create vite@latest`

2. Select

- Framework → React
- Variant → JavaScript

3. Install dependencies

Command: `npm install`

4. Run the project

Command: `npm run dev`

The development server starts successfully and the project opens in the browser.

```
Need to install the following packages:
create-vite@9.1.1
Ok to proceed? (y) y
```

```
> npx
> create-vite
```

```
◇ Project name:
  Task2
◇ Package name:
  task2
◇ Select a framework:
  React
```

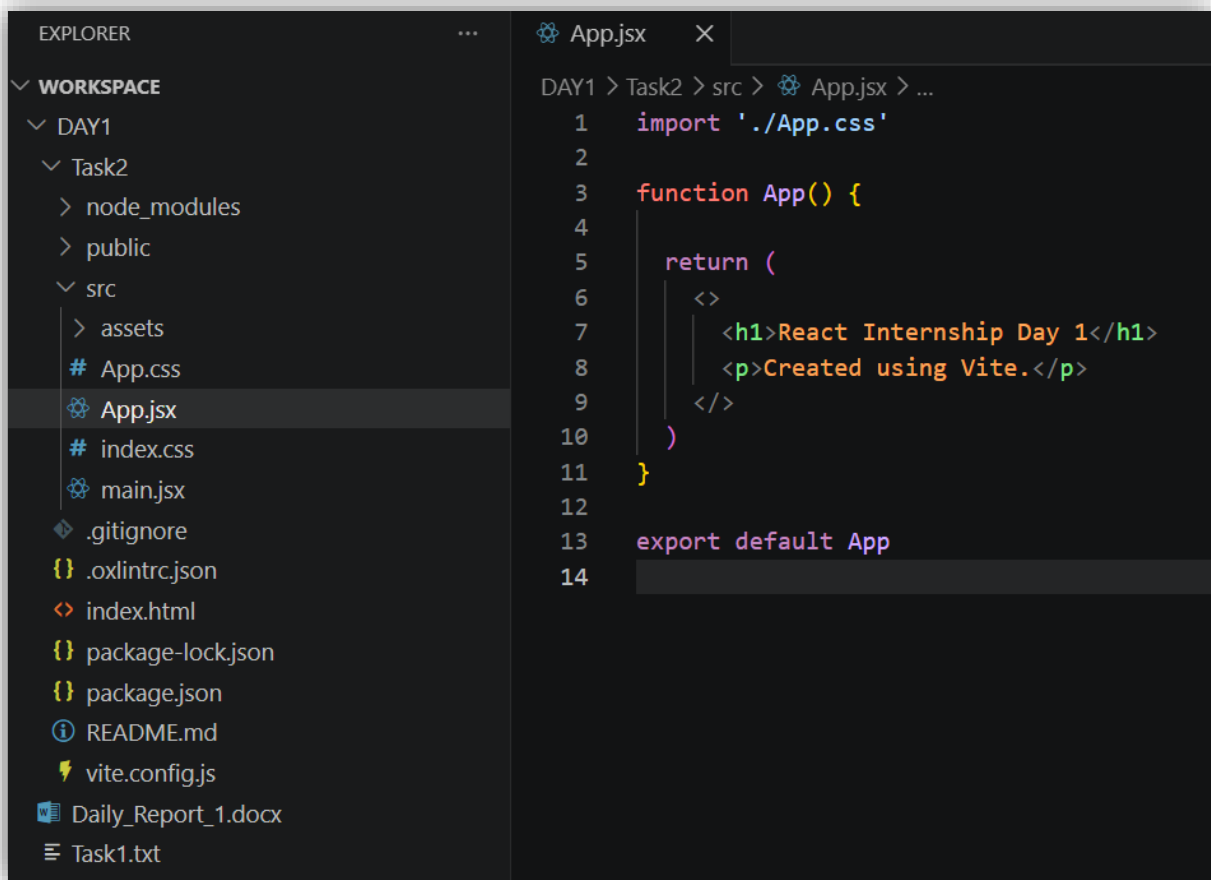
```
◇ Project name:
  Task2
◇ Package name:
  task2
◇ Select a framework:
  React
◇ Select a variant:
  JavaScript
◇ Which linter to use?
  Oxlint
◇ Install with npm and start now?
  Yes
```

```
◇ Starting dev server...

> task2@0.0.0 dev
> vite

VITE v8.1.3 ready in 1518 ms

→ Local:   http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
```



The screenshot shows the Visual Studio Code interface. On the left, the Explorer sidebar displays the project structure under 'WORKSPACE'. The 'src' directory is expanded, showing files like 'App.css', 'App.jsx', 'index.css', 'main.jsx', and others. The 'App.jsx' file is selected. The main editor area shows the content of 'App.jsx', which includes an import for './App.css', a function 'App()' that returns a JSX element with an h1 and a p, and an export default statement.

```
App.jsx
1  import './App.css'
2
3  function App() {
4
5      return (
6          <>
7              <h1>React Internship Day 1</h1>
8              <p>Created using Vite.</p>
9          </>
10     )
11 }
12
13 export default App
14
```

React Internship Day 1

Created using Vite.

Summary

Important Files in a React Project:

1. App.jsx

- Main component of the application.
- Most of the React code is written here.
- It controls the UI of the application.

2. main.jsx

- Entry point of the React application.
- Renders the App component inside the root element.

Example:

```
createRoot(document.getElementById("root")).render(  
  <App />  
);
```

3. index.css

- Contains global CSS styles.
- Applied to the whole application.

4. App.css

- Contains styles specific to the App component.

5.index.html

The **index.html** file contains

```
<div id="root"></div>
```

React renders the entire application inside this **root div**.

=> JSX Rules:

- Return only one root element.
- Use fragments (<> </>) if multiple elements are required.
- Tags must always be closed.
- JSX is case-sensitive.
- Components should start with a capital letter.

Example using Fragment:

```
<>
  <h1>Hello</h1>
  <p>Welcome</p>
</>
```

Fragments avoid adding unnecessary <div> elements to the DOM.

Import and Export

React uses ES6 modules.

Default Export:

```
export default App;
```

Import:

```
import App from "./App";
```

Named Export:

```
export { Title };
```

Import:

```
import { Title } from "./Title";
```

JavaScript inside JSX

JavaScript expressions can be written inside curly braces {}.

Example:

1. `const msg = "Hello";`

`<p>{msg}</p>`

Output: Hello

2. `<p>{2 * 2}</p>`

Output: 4